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Canadian Cordilleran displacement: palaeomagnetic results from the Early Jurassic Hazelton Group, Terrane I, British Columbia, Canada

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SUMMARY

The Early Jurassic Telkwa Formation, comprising the base of the Hazelton Group, was sampled at three areas in the central part of the Stikine Terrane, British Columbia. Detailed alternating field and thermal step demagnetization on samples from nine sites from the Red Canyon area and 10 sites from the Zymoetz River area, both of which are in the Bulkley Ranges, and eight sites from the Telkwa Range area isolate stable remanence magnetization components. The components are interpreted to be primary magnetizations because of the presence of two polarities and the much improved agreement of their inclinations after full tilt correction of the sites from the Bulkley Ranges. Within-locality declinations agree well; between-locality declinations are discordant. The expected cratonic declination is ~340° whereas declinations from this study are 265°, 227° and 328° for the Red Canyon, Zymoetz River and Telkwa Range areas respectively. These results are similar to those of Monger & Irving (1980) and we concur that the discordant declinations are the result of differential rotation about vertical axes between sample localities. Our observed mean inclination of 52° is in good agreement with Monger & Irving's (1980) result of 54°. They interpreted this to indicate ~1300 km of northward displacement since the Early Jurassic and their interpretation was substantiated by several later studies. However, with revisions to the geologic time-scale and the Early Jurassic reference pole for North America, the Early Jurassic Canadian Cordillera palaeomagnetic data base appears latitudinally concordant. This indicates that Terrane I and southern Wrangellia were in much the same latitudinal position relative to the North American craton as they are now. Tectonic displacement models developed in view of discordant Cretaceous palaeomagnetic data from the Canadian Cordillera must now consider latitudinally concordant Permian to Early Jurassic data and Middle Eocene to Recent data. This tectonic senario appears more reconcilable with the regional tilt hypothesis. However, this hypothesis cannot account for discordant Cretaceous data from bedded volcanic rocks in Terrane I. It is therefore probable that Cretaceous Cordilleran tectonics were characterized by both regional tilting and moderate latitudinal displacement.

Key words: Cordilleran tectonics, Early Jurassic, Hazelton Group palaeomagnetism, Telkwa Formation, Terrane I.

1 INTRODUCTION

The tectonic development of the Canadian Cordillera reflects the complex evolution of a collage of allochthonous terranes. Palaeomagnetism has been useful in studying this development by providing quantitative analyses of large-scale tectonic motions within and between terranes. The first aberrant palaeomagnetic result in the western Cordillera of North America came from the Eocene Siletz River Volcanic series of Oregon (Cox 1957). However, it was not until later that Irving (1964) recognized that this aberrant result was caused by rotation about a vertical axis. Since then numerous discordant palaeomagnetic results, together with differences between Cordilleran and contiguous North American stratigraphies and differences between coeval faunas along the ancient cratonic margin, have led to the recognition that the Cordillera is a collage of terranes of different origin. This collage of allochthonous terranes

assembled and collided with the ancient cratonic margin at different times.

There are numerous published palaeomagnetic studies with which to test for large-scale motions in the Canadian Cordillera. However, considerable controversy exists over the interpretation of discordant palaeomagnetic data from the Early Jurassic to mid-Cretaceous (e.g. Monger & Irving 1980; May & Butler 1986; Butler *et al.* 1989). It involves different perceptions of the geologic development of the Cordillera and differences in the analytical procedures used to calculate North American reference poles. The purpose of this study is to examine this controversy by a review of the data base with the addition of new palaeomagnetic data from the Early Jurassic Telkwa Formation, a volcanic suite from the Stikine Subterrane of Terrane I (Monger, Price & Tempelman-Kluit 1982; Monger 1984) (Fig. 1).

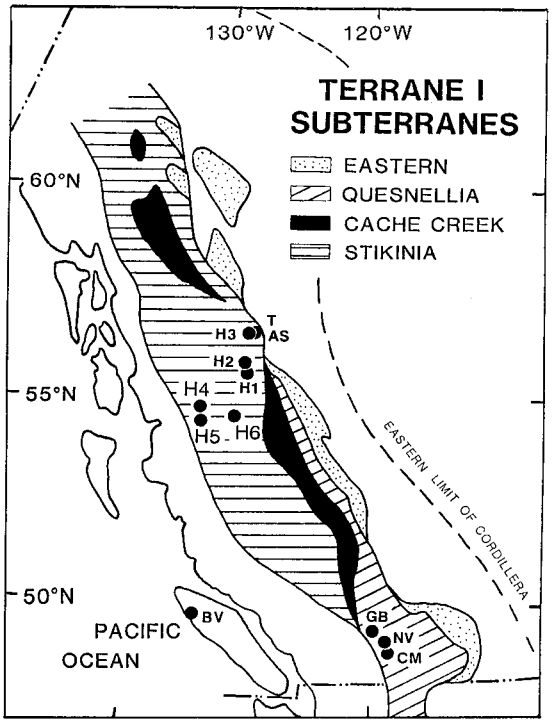


Figure 1. Terrane map with the study areas for the Hazelton Group [H1–H3, Monger & Irving (1980), and H4–H6, this study], Takla Group (T), Asitka Group (AS), Guichon Batholith (GB), Nicola Group (NV), Copper Mountain Intrusions (CM) and Bonanza Group (BV), in relation to the subterranes of Terrane I from Monger *et al.* (1982).

The Telkwa Formation is part of the Hazelton Group and was selected for this study because it has been biochronologically dated, crops out over an extensive region of the central Stikine Terrane, and has bedding attitudes to define the palaeohorizontal unlike plutonic rocks (e.g. Beck Burmester & Schoonover 1981; Irving *et al.* 1985). Thus it is possible to relate the directions of remanent magnetizations to the estimated palaeohorizontal when the primary remanence is acquired.

Previous palaeomagnetic work on the volcanic rocks of the Hazelton Group by Monger & Irving (1980) was restricted to areas near the eastern margin of Stikinia (H1, H2, H3; Fig. 1). Their results suggested that Stikinia had been translated northward by ~1300 km as well as rotated substantially about vertical axes. This interpretation was based on the best available reference pole at that time (Irving 1979). Although, the precision of their Hazelton Group result is somewhat low, the interpretation was supported by results from the Takla Group and Axelgold intrusion (Monger & Irving 1980). Conclusions about northward displacement and large-scale variable rotations for parts of the Canadian Cordillera from several later studies on Early Jurassic rocks (e.g. Symons 1983; Symons & Litalien 1984; Symons 1985; Irving & Yole 1987) were consistent with Monger & Irving's (1980) result. However, these conclusions involved the use of different Early Jurassic North American reference poles. Thus, the proposed displacements and rotations were not calibrated with one another, and therefore, should not be compared. In addition, the North American Early Jurassic reference pole was imprecisely defined. This is demonstrated by numerous revisions over the last decade (e.g. Irving 1979; Irving & Irving 1982; Gordon, Cox & O'Hare 1984; May & Butler 1986; Irving & Yole 1987; Hodych & Hayatsu 1988). Unless the reference pole is accurately known within a very few degrees of error a tectonic analysis will be similarly inaccurate. Latitude displacement errors can be of the order of several hundred kilometres and rotation errors of several tens of degrees, both of which can be independent of the precision of the pole determination. Thus, in order to accurately evaluate the tectonic implications of these earlier palaeomagnetic studies of Early Jurassic rocks the tectonic displacements must be recalculated using the same, current, more precisely defined, Early Jurassic reference pole.

Recent reviews of the Early Jurassic palaeomagnetic data, which include Monger & Irving's (1980) work on the Hazelton Group, by May & Butler (1986) and Vandall, Palmer & Woodsworth (1989) suggest that originally proposed large-scale northward displacements of the Canadian Cordillera are no longer supported by current cratonic reference poles. This issue is addressed in this study along with the consequences for tectonic models developed in view of discordant Cretaceous palaeomagnetic data.

2 GEOLOGY AND SAMPLING

The Early to mid-Jurassic Hazelton Group (Leach 1910) is a continental to island-arc calcalkaline assemblage that was deposited in the northwest-trending Hazelton trough (MacIntyre & Desjardins 1988). Tipper & Richards (1976) divided it into three major formations that are, in ascending stratigraphic order, the Telkwa Formation, the Nilkitkwa Formation and the Smithers Formation. The Telkwa Formation, of Late Sinemurian to Early Pliensbachian age (206–198 Myr; Harland *et al.* 1982), crops out in mid-central Stikinia and comprises the thickest and most extensive formation of the Hazelton Group. It includes subaerial to submarine pyroclastic and flow rocks with lesser intercalated sedimentary rocks (MacIntyre & Desjardins 1988). Telkwa Formation volcanic units were sampled in the Red Canyon area (H4), Zymoetz River area (H5) and Telkwa Range (H6) (Figs 1 and 2).

The Red Canyon area (Fig. 2, H4) is characterized by a moderately shallow east–northeast dipping homoclinal succession of greenish-grey calcareous tuffs and red crystal

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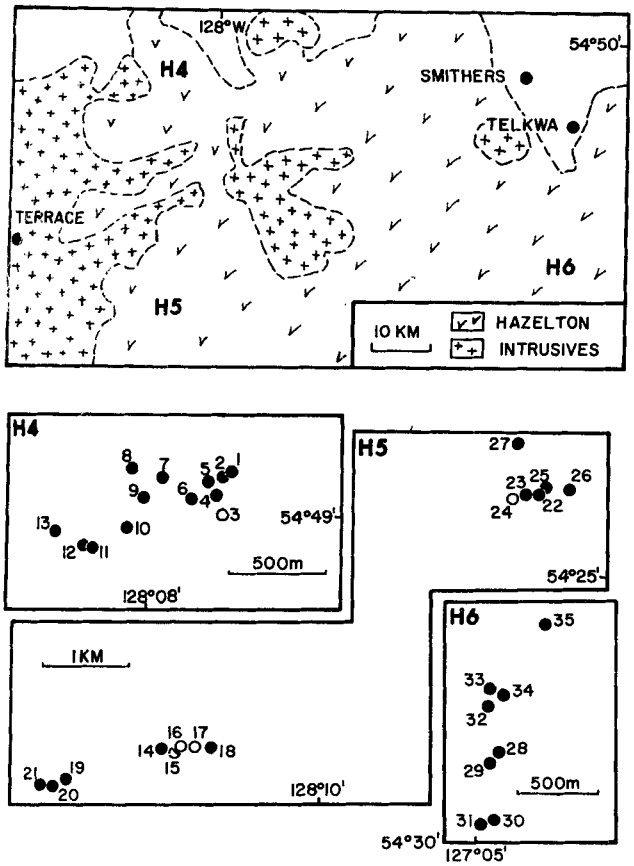


Figure 2. Regional setting of the Hazelton Group sample areas in mid-central Stikinia. Base map modified from Tipper, Woodsworth & Gabrielse (1981). Sample areas, H4 Red Canyon sites 1–13, H5 Zymoetz River sites 14–27 and H6 Telwa Range sites 28–35. Solid circles are normal polarity sites and open circles are sites with reverse polarity.

tuffs. The top of the ≥400 m sampled section contains Sinemurian fossils and is just below the Smithers Formation (Woodsworth, Hill & van der Heyden 1985). It is reasonably well exposed so that accurate bedding attitudes could be measured at or near the 13 sample sites. Most of these sites were sampled in the red crystal tuffs. The metamorphic grade appears to be very low as detailed studies of the adjacent Zymoetz River area (Mihalynuk 1987) and the Telkwa Range area (Dudley 1983) have indicated that regional metamorphism grades from zeolite to prehnite–pumpellyite facies. In addition, the area is far removed from known intrusive bodies (Woodsworth *et al.* 1985), and therefore, it is not likely that it has experienced any thermal remagnetization events since deposition.

The Zymoetz River area (Fig. 2, H5) is characterized by a moderately steep eastward-dipping homoclinical succession composed of intercalated bladed-feldspar porphyry flows, crystal tuffs and a mauve ignimbrite that crops out as a conspicuous marker horizon (Mihalynuk & Ghent 1986). The succession is exposed for 25 km in the down dip direction but some of the section is repeated by thrust faulting so that it probably represents about 6–7 km of Telkwa stratigraphy (Mihalynuk & Ghent 1986). West of the Zymoetz River area, the Telkwa Formation rocks have been intruded by the Coast Plutonic Complex but even there the higher metamorphic grades are restricted to narrow contact zones (Mihalynuk 1987). In the Zymoetz River area, 14 sites were sampled that represent more than 1 km of stratigraphic section. The exposures are very good so that bedding attitudes are easily determined.

The rocks of the Telkwa Formation in the Telkwa Range, southwest of Smithers (Fig. 2, H6), are dominantly lava flows (MacIntyre *et al.* 1989). Aphyric basalts were sampled at eight sites from a gently northward-dipping sequence with Sinemurian fossils in overlying sediments (MacIntyre *et al.* 1989). The eight sites are distributed over about 200 m of section. Again the bedding attitudes could be readily determined and most of the sample sites were subhorizontal. The regional metamorphic grade is zeolite facies (Dudley 1983) so that these flows are the most pristine Telkwa Formation rocks studied to date. The area is not known to have been intruded by later igneous rocks (MacIntyre *et al.* 1989).

Sinemurian fossil localities immediately above two of the sampled sections indicate an age of 200–206 Myr on the geologic time-scale of Harland *et al.* (1982).

3 METHODS

In total, 35 palaeomagnetic sites in the Telkwa Formation were collected in flows and tuffs using a portable drill. At least five cores were drilled and oriented by magnetic and/or solar compass at each site. In the laboratory two specimens were cut from each core. The specimens were analysed using a Schonstedt DSM-1 spinner magnetometer, GSD-5 tumbling alternating field demagnetizer and TSD-1 thermal demagnetizer. After measuring the natural remanent magnetization (NRM), each specimen was subjected to detailed alternating field (AF) and/or thermal stepwise demagnetization in up to 12 steps. Alternating field demagnetization was done up to maximum fields of 100 mT and thermal demagnetization was done up to peak temperatures of 700 °C. Characteristic remanence directions in each specimen were obtained by a joint analysis of stereographic and orthogonal demagnetization diagrams and least-squares principal-component analysis following the method of Kirschvink (1980). Remanence directions were accepted with a mean angular deviation (MAD) < 10°. The site mean remanence directions were calculated using conventional tiered statistics (Fisher 1953; Irving 1964).

4 RESULTS

The majority of specimens exhibit stable linear vector decay (Fig. 3a–f). After the initial removal of low coercivity and low unblocking temperature viscous components (VRM), the majority of specimens decayed substantially and linearly towards the origin with maximum alternating fields of 100 mT (Fig. 3d–3f) and/or thermal unblocking temperatures typical of both magnetite and haematite (Fig. 3a–c), showing that the same stable remanence direction is carried by both magnetite and haematite. The VRM directions from one specimen to the next were quite different indicating randomly directed viscous components some of which were acquired during a storage test in the laboratory. The data from four Red Canyon sites (1, 2, 4 and 7) and four Zymoetz River sites (14, 19, 25, and 26)

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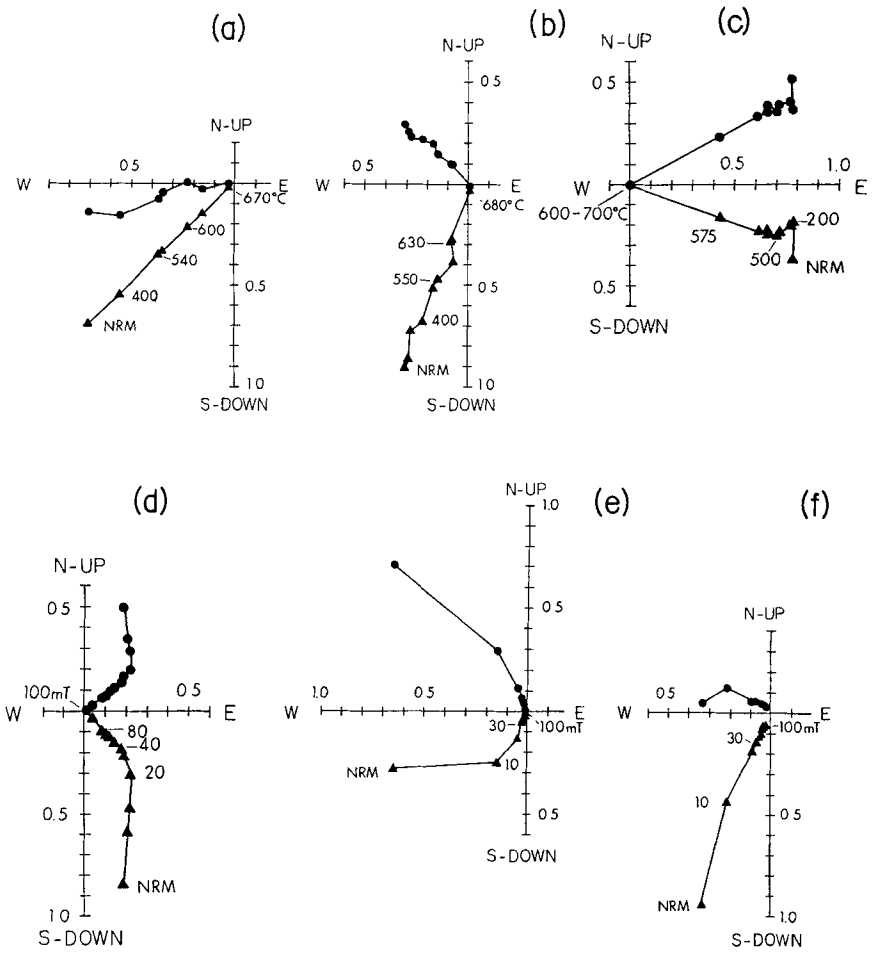


Figure 3. Orthogonal vector decay plots for example specimens on thermal and AF step demagnetization. Solid circles represent vector projections on the horizontal plane and solid triangles, vector projections on the vertical plane; all symbols denote down directions; the AF step intensities are in millitesla (mT); and the thermal step temperatures are in degrees Celsius (°C). The axial values are given as a ratio of the NRM intensity. The directions are before tilt correction. Specimens are from sites 9 (a), 13 (b), 17 (c), 17 (d), 29 (e) and 31 (f). Note specimens (c) and (d) from site 19 are reverse polarity after tilt correction.

were rejected because of unstable magnetic remanence, attributable to either very weak magnetic intensities of the order of 0.1 mA m^{-1} in some sites or to anomalously strong magnetic intensities in other sites that are indicative of lightning-induced isothermal remanence. Thus stable remanence directions were isolated in 27 of the 35 sites, and their corresponding site means and area means are shown in Tables 1 and 2.

Only downward directed magnetizations were isolated in the Telkwa Range and Red Canyon sites (Table 1; Fig. 4a) except for site 3 where both upward and downward directed vectors were observed in different layers of a thinly laminated tuff. The *in situ* site mean directions from the rocks in the Telkwa Range area where structural attitudes are subhorizontal form a reasonably tight cluster with a north–west directed moderately steep down direction (Fig. 4a) which is close to the expected Early Jurassic direction as calculated from results from the craton. In contrast, the *in situ* site mean westward-directed magnetizations from the Red Canyon area display greater scatter, specifically in declination; the shallow inclinations are reasonably well grouped. In the Zymoetz River area, both upward and downward directed *in situ* shallow magnetizations are roughly antipodal to the northeast and down, and southwest and up (Fig. 4a). Within a sample area, the site mean directions agree reasonably well before tilt correction. However, between sample areas the declinations and inclinations are quite scattered and in particular the directions from the Zymoetz River and Red Canyon areas are well removed from the expected Early Jurassic direction (Fig. 4a).

4.1 Tilt test

Given the very shallow subhorizontal attitude of the basalts of the Telkwa Range it is quite possible that the shallow northerly dip is primary and reflects a gently sloping surface away from a volcanic centre. This would suggest that the mean remanence direction for the Telkwa Range area should be closer to the expected cratonic reference pole before tilt correction than after and this is in fact the case (Table 2) but the differences are not statistically significant.

In contrast, the remanence data from the Red Canyon and Zymoetz River areas, hereafter referred to jointly as

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Table 1. Site mean remanence data.

Site	Area	N	D°	I°	ϕ°	ψ°	CD°	CI°	R	k	α95°
3	H4	6	288	38	099	24	293	62	5.599	13	20 *
5	"	2	288	43	099	24	295	66	1.827	–	–
6	"	7	280	27	099	24	281	51	6.862	43	9
8	"	7	248	28	095	25	237	50	6.859	43	9
9	"	9	244	32	095	25	230	52	8.748	32	9
10	"	8	278	20	095	25	279	45	7.818	39	9
11	"	8	238	23	075	28	231	49	7.883	60	7
12	"	9	265	32	075	28	272	60	8.865	60	7
13	"	3	274	30	075	28	285	55	2.776	9	44
15	H5	10	059	5	080	70	038	–57	9.810	48	7
16	"	9	048	7	080	70	027	–49	8.700	27	10
17	"	7	054	21	080	60	051	–34	6.848	39	10
18	"	5	263	–8	105	74	241	58	4.785	19	18
20	"	8	225	–20	069	74	215	47	7.789	33	10
21	"	6	230	–16	071	75	216	53	5.992	592	3
22	"	5	235	–17	090	70	225	40	4.815	22	17
23	"	3	259	2	090	70	237	69	2.684	6	53
24	"	5	073	21	090	70	067	–46	4.933	60	10
27	"	7	247	–15	099	70	235	43	6.910	66	8
28	H6	7	334	41	000	3	335	39	6.875	47	9
29	"	6	316	51	000	3	318	49	5.916	60	9
30	"	5	344	44	000	0	344	44	4.854	27	15
31	"	8	319	54	000	0	319	54	7.745	28	11
32	"	7	310	41	000	5	313	38	6.969	197	4
33	"	5	350	51	000	5	351	46	4.647	16	24
34	"	5	323	53	000	5	326	49	4.646	12	24
35	"	5	315	54	000	5	319	51	4.897	39	13

Notes: N is the number of samples; D and I are the mean remanence vector's declination and inclination, respectively, relative to present horizontal; ϕ is the down dip azimuth of bedding and ψ is the dip angle from horizontal; CD and CI are the the same as D and I but are after tilt correction relative to bedding; R is the resultant vector for the summation; k and α95 are the precision parameter and radius of the circle of 95% confidence (Fisher 1953). H4–Red Canyon Area, H5–Zymoetz River Area, H6–Telkwa Range Area. * – Site 3 is characterized by 3 normal polarity and 3 antipodal reverse polarity specimens.

the Bulkley Ranges, come from much steeper dipping layers. The layers include thinly bedded airfall tuffs that were presumably deposited sub-horizontally and initial dip is not likely to be a problem. In the Bulkley Ranges the rocks come from homoclinal successions so that the within-area tilt corrections are similar for all sites and the within-area fold test (Graham 1949) is somewhat inconclusive. Thus the precision parameter (k) (Fisher 1953) for the Red Canyon area improves only marginally upon untilting (Table 2). Untilting of the directions from the steeply dipping section,

Table 2. Uncorrected and corrected area means and their palaeopoles.

Area	U/C	N	D°	I°	k	α95°	Dp°	Dm°	Long°	Lat°
H4	U	9	266	32	21	11.5	7	13	155E	12N
"	C	9	265	56	24	10.7	11	15	169E	27N
H5	U	10	241	–12.9	28	9.3	5	10	162E	22N
"	C	10	227	50	38	8.0	7	11	013E	5N
H6	U	8	328	50	51	7.9	7	11	112E	56N
"	C	8	326	47	53	7.7	6	10	107E	55N

Notes: U corresponds to an uncorrected mean, C corresponds to a tilt corrected mean. N is the number of site means. D, I, k, α95 are same as in Table 1. Dp is the radius of the ellipse of 95 per cent confidence at the pole along the site–pole great circle and Dm is the radius of this ellipse perpendicular to the site–pole great circle. Long is the pole longitude and Lat is the pole latitude. H4, H5 and H6 same as in Table 1.

the Zymoetz River section, does however improve the clustering of the mixed polarity data (Fig. 4b) with the precision parameter (k) (Fisher 1953) increasing from 28 to 38 (Table 2). On comparing the directions from all three localities, before and after full tilt corrections have been applied to the data from the Bulkley Ranges, the fold test is more conclusive in that the inclinations agree well after correction even though the declinations do not (Fig. 4b). The much improved agreement of site mean inclinations and the improved within-area precision of the Zymoetz River

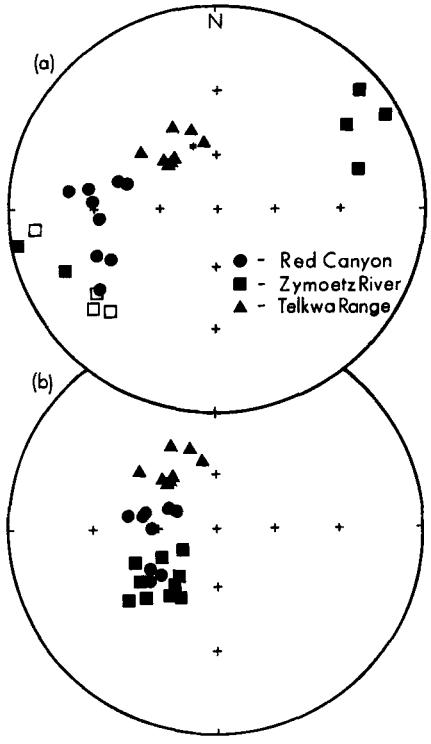


Figure 4. Equal-area lower hemisphere stereographic projection showing site mean directions (a) present horizontal where closed symbols are normal vectors and open symbols are reversed vectors and (b) the same but tilt corrected relative to bedding with five reverse sites plotted in their antipodal normal position.

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area after tilt correction strongly suggest a positive fold test for the Bulkley Ranges. The declination differences are thought to reflect large rotations about vertical axes between the sample areas. To derive a precision estimate of the tilt test the magnitudes of rotations between areas were determined by calculating the angles between the observed mean declinations and the expected field declination and then applying these rotation angles to the individual site mean directions. The Telkwa unit mean was calculated then before and after tilt correction, resulting in an improvement of the precision parameter k from 7 to 35, respectively. This corresponds to a positive fold test passing at well above the 99 per cent confidence level (McElhinny 1964; McFadden & Jones 1981).

4.2 Magnetostratigraphy

Five of the 27 sites with stable magnetic remanence have reverse polarity magnetizations. One of these five sites in delicately laminated tuffs in the Red Canyon area has both normal and reverse magnetizations. The single mixed polarity site (3) is the uppermost stably magnetized site in the Red Canyon sequence (Fig. 2, H4). The other four reversed sites are from the bases of two sections separated by 5 km of poorly exposed and unsampled volcanic rocks within the Zymoetz River area (sites 15–17 and 24; Fig. 2, H5). Reversed polarity sites 15, 16 and 17 are stratigraphically ordered in successive units upwards within a ~200 m section. Reversed site 24 is overlain by normal polarity volcanic lava flows. Therefore at least one normal and one reversed polarity chron is recorded in both areas. As correlation between the areas is uncertain and duplication of stratigraphy by faulting is possible, suspected additional polarity chrons cannot be confirmed. It is known that the Early Jurassic was a period of many polarity reversals (Steiner & Ogg 1988), so that rocks of this age which retain a primary remanence can be expected to record polarity reversals. The isolation of a stratigraphically zoned magnetization with antipodal directions is good evidence that the remanence is primary and Sinemurian to Early Pliensbachian in age.

5 DISCUSSION

5.1 Hazelton Group

The stable magnetizations isolated in the Telkwa Formation rocks are characterized by high unblocking temperatures and high coercivities. In two of the three sample areas antipodal magnetic reversals are recorded. The tilt test, after rotation correction about vertical axes, is positive and indicates that the remanence from the Bulkley Ranges is pre-tilting at >99 per cent confidence. Therefore we interpret this to be a primary magnetization that was acquired upon cooling of the lava flows and deposition of the tuffs.

A characteristic feature of the Hazelton Group volcanic rocks sampled in Monger & Irving's (1980) study was the large apparent rotations between sampling areas. We attempted to minimize these effects by sampling farther from the Stikine Terrane boundaries where 'ball-bearing type' terrane margin rotations are probable (Beck 1980).

Table 3. Summary of Hazelton Group palaeomagnetic results.

Sample Area	N	D°	I°	k	α95°	Polarity	Mean I°
Results from Monger and Irving (1980)							
H1	7	359	55	15	16.0	mixed	
H2	4	242	56	29	18.0	normal	
H3	4	114	–52	14	25.0	reversed	54
Results from this study							
H4	9	265	56	24	10.7	mixed	
H5	10	227	50	38	8.0	mixed	
H6	8	328	50	51	7.9	normal	52

Notes: N, D, I, k, α95, H4, H5, H6 are same as in Table 2.
H1 – Bait Range Area, H2 – Driftwood Range Area, and
H3 – Red Creek–Two Lake Creek Area (Monger and Irving 1980).

However, our three ample areas are within large fault bounded blocks that have also been rotated but to a lesser extent. The central location of the Telkwa Range is likely to have undergone little 'ball-bearing type' terrane margin rotation, and this is confirmed by the close agreement of its mean direction with the expected cratonic direction.

Our results are similar to those of Monger & Irving (1980) for the Hazelton Group (Table 3, Fig. 5). They also sampled the Hazelton Group in three areas (H1, H2, H3; Fig. 1) and discovered large variations in declination not only between sample areas but also within sample areas. Their area mean precisions were low, partly due to the declination scatter but also due to the low number of sites. This study substantiates, with improved precision, Monger & Irving's (1980) palaeomagnetic results and demonstrates that large block rotations about vertical axes have taken place on both the eastern and western margins of the Stikine Terrane since the Early Jurassic. Central Stikinia, represented by our Telkwa Range result, has apparently undergone relatively

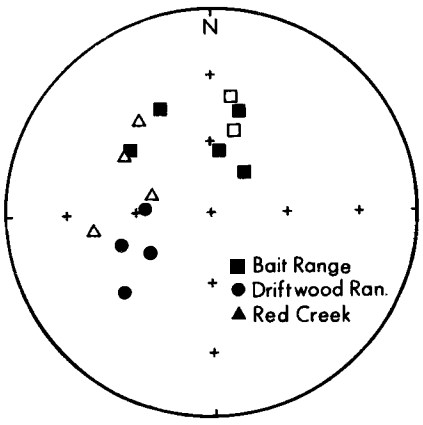


Figure 5. Equal-area stereographic projection showing the site mean directions which have been tilt corrected relative to bedding from Monger & Irving (1980). Closed symbols are normal vectors and open symbols are reverse vectors which have been plotted in their antipodal normal position.

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little rotation. Assuming that rotation has been in the smallest angle sense, then post-Jurassic counterclockwise rotations have been observed on the west side of Stikinia in this study. In contrast, both counterclockwise and clockwise rotation have been observed on the east side of Stikinia by Monger & Irving (1980). This pattern suggests these localized rotations did not affect Stikinia as a whole, but are confined to areas proximal to the terrane margins where accretion tectonics on both sides of Stikinia could generate the block rotations.

5.2 Review of Early Jurassic data

Using the reference poles for cratonic North America of Irving (1979), Monger & Irving (1980) found that their poles from the three Hazelton Group sections, together with the Triassic Takla Group and the Cretaceous Axelgold intrusion poles, were discordant. Their tectonic analysis suggested a northward displacement of ~1300 km for Stikinia had taken place since the units had acquired their magnetizations. Results from several later studies on rocks within Terrane I and Wrangellia were consistent with the northward displacement hypothesis, including the Guichon batholith (Symons 1983; GB Fig. 1), Copper Mountain intrusions (Symons & Litalien 1984; CM), Nicola volcanics (Symons 1985; NV) and the Bonanza Group (Irving & Yole 1987; BV). Gordon *et al.* (1984) pointed out that Monger & Irving's (1980) interpretation, although based not only on palaeomagnetic evidence, but also on palaeontological and geological evidence, was not consistent with their updated Late Triassic to Early Jurassic reference poles. These poles were calculated by the palaeomagnetic Euler pole (PEP) analysis (Gordon *et al.* 1984) using the geological time-scale of Harland *et al.* (1982). They indicated a smaller northward displacement for the Takla Group and a southward displacement for the Hazelton Group. Although we question the PEP procedure of Gordon *et al.* (1984), as will be discussed, this work illustrated the need for a new tectonic analysis of the palaeomagnetic results in view of: the significantly revised geologic time-scale, the addition of new palaeomagnetic results to the North American data base, and the recent radiometric dating of palaeomagnetically studied rock units.

A recent tectonic analysis was carried out by May & Butler (1986) who recalculated the Jurassic apparent polar wander (APW) for North America. They also used the PEP analysis and reconsideration of the palaeomagnetic data from Terrane I led to the conclusion that Terrane I was in much the same position relative to North America as it is now. They also pointed out that owing to the poor precision of the Hazelton Group remanence data (Monger & Irving 1980), latitudinal resolution of the Hazelton results was limited. In as much as the data of this study substantiate the results of Monger & Irving (1980) but with greater precision, increased latitudinal resolution is now possible.

Three analytical procedures have been used to calculate reference poles for use in tectonic analyses. The earliest procedure involves a time-window moving average technique using intervals of equal time within which poles are averaged. This technique was used by Irving (1979), Harrison & Lindh (1982) and Irving & Irving (1982), typically with a 30 Myr time-window. Although this procedure is a very useful technique, it can only, at present, illustrate the first-order APW as it is limited by an incomplete and possibly time-biased palaeomagnetic data base, and therefore, tends to smooth out polar wander details especially in the vicinity of cusps. However, as the data base increases and becomes more complete much smaller time windows can be used to eliminate the undesirable smoothing of APW.

By another procedure, palaeomagnetic Euler poles (PEP) (Gordon *et al.* 1984) are calculated from palaeopoles that are fitted to small circles that are thought to represent a plate trajectory that remained constant over some 100 Myr. This technique is constrained by a tectonic model which assumes that plate trajectories do not change more frequently than every 100 Myr or so. However such changes can occur more frequently (Engbretson 1982). Poles over a range of ~100 Ma fitted to a small circle assume that information about the Triassic magnetic field direction, for example, can be obtained from the Late Jurassic magnetic field direction. This is a pre-judgment of the nature of APW being investigated.

The third technique (e.g. Irving & Yole 1987; Hodych & Hayatsu 1988) is very simple, relying directly on observations of the ancient magnetic field direction. One simply selects from the current cratonic data base, in our case North America, well-defined and dated cratonic palaeomagnetic poles of the relevant age within the narrowest possible time-window. These poles are then averaged giving each pole unit weight, to obtain a reference pole. This procedure is similar to the time-window moving average technique except that the size of the window is optimized (e.g. 5–10 Myr), centred about the mean age of the rock unit being studied. We agree with Irving & Yole (1987) that this is the most prudent, practical and accurate way of obtaining a reference pole suitable for use in tectonic analyses.

The ages of the Early Jurassic magnetizations from Terrane I and Wrangellia (Monger & Irving 1980; Symons 1983; Symons & Litalien 1984; Symons 1985; Irving & Yole 1987; this study) are ~200 Myr, using the geologic time-scale of Harland *et al.* (1982). Therefore, an accurate Early Jurassic North American cratonic reference pole is required for a tectonic analysis of these palaeomagnetic results. Irving & Yole (1987) calculated an Early Jurassic reference pole for their tectonic analysis of palaeomagnetic data from the Bonanza Group in Wrangellia on Vancouver Island. Their reference pole has its weaknesses. It is based on relatively few poles (6) from cratonic North America as data from the Colorado Plateau were omitted. Three of the six accepted poles are from intrusives which likely represent spot readings of the geomagnetic field. In addition, it includes poles (4) that were rotated into the North American reference frame from the African craton. We prefer the use of North American data, only to avoid the added complications that may be introduced when poles from other cratons are rotated into the North American reference frame. Prévot & McWilliams (1989) discuss problems with the Early Jurassic reference pole for North America and suggest it is possible that little geomagnetic secular variation has been recorded by each of the extrusive and intrusive units in the Mesozoic basins of eastern North

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America and caution the weight assigned to them in tectonic interpretations.

It has been recognized by several workers that the Colorado Plateau has undergone clockwise rotation, suggested by both geologic (Hamilton 1981) and palaeomagnetic evidence (Steiner 1984). Recent estimates on the amount of clockwise rotation of the Colorado Plateau include: 4° (Bryan & Gordon 1986), 11° (Steiner 1986) and 10° (Irving & Yole 1987). Although Irving & Yole (1987) dismiss data from the Colorado Plateau on several grounds, their strongest argument is the rotation complication. Our approach is different. Much of the Early Jurassic Cordilleran data is characterized by large and variable declination differences so that the most useful information comes from their inclinations which are directly related to palaeolatitudes. Correcting the data from the Colorado Plateau for clockwise rotation of either 4° or 11° has a small effect on the longitude of a combined Colorado–Newark Early Jurassic reference pole and a negligible effect on its latitude. Since inclusion of these data does not change the expected cratonic reference latitude but does help to average out secular variation, we argue that a combined Colorado–Newark North American reference pole best averages out secular variation, and therefore, is the most accurate for the needs of this analysis.

Recently Hodych & Hayatsu (1988) have reviewed the age dating and palaeomagnetic work on the Newark Supergroup rocks that provide most of the Early Jurassic cratonic poles and have calculated an Early Jurassic reference pole. We used nine poles from the Newark Supergroup data combined with three poles from the Colorado Plateau corrected for 4° of clockwise rotation to calculate our Early Jurassic reference pole for North America (Table 4). A comparison of Early Jurassic reference poles (Table 4) shows how closely some of the various independant calculations agree in spite of the inclusion of different data.

Table 4. Early Jurassic reference poles for cratonic North America.

	Long°	Lat°	A95°	N	Reference
	64 N	94 E	2.8	16	Harrison and Lindh (1982)
	63 N	92 E	5.0	12	Irving and Irving (1982)
a	68 N	95 E	5.0	6	Irving and Yole (1987)
b	64 N	88 E	8.0	3	Irving and Yole (1987)
c	64 N	87 E	2.3	24 *	Smith and Noltimer (1979)
d	64 N	85 E	6.1	9	Hodych and Hayatsu (1988)
e	63 N	85 E	7.0	12	Combined Colorado–Newark

Notes: Long, Lat and A95 same as before. *N* is the number of poles averaged. * is the number of site poles averaged. a) Average of their selected North American Poles. b) Average of their North American Poles omitting those poles believed not to have averaged secular variation. c) Combined groups I and II of Smith and Noltimier (1979). d) Includes poles from the Newark rocks and equivalents. e) Combined Colorado–Newark pole of this study includes nine poles from the Newark rocks and equivalents from Hodych and Hayatsu (1988) and three Colorado Plateau poles used by May and Butler (1986) corrected for 4° of clockwise rotation.

Table 5. Tectonic analysis.

Study	Ref. Pole	RPD°	RR°	Reference
Hazelton	a	0 ± 10	+14, +79, +117	(This study)*
"	b	−2 ± 8	+14, +78, +117	
Hazelton	a	−2 ± 17	−15, +50, +102	(Monger and Irving 1980)*
"	b	−3 ± 17	−15, +50, +102	
Takla	a	2 ± 5	+42	(Monger and Irving 1980)
"	b	5 ± 5	+61	
Guichon	a	6 ± 9	−41	(Symons 1983)
"	b	5 ± 9	−41	
Nicola	a	−3 ± 9	−53	(Symons 1985)
"	b	−4 ± 9	−53	
Copper	a	1 ± 7	−37	(Symons and Litalien 1984)
"	b	1 ± 7	−37	
Bonanza	a	3 ± 9	+69	(Irving and Yole 1987)
"	b	2 ± 9	+68	

Notes: reference poles (a) 64N, 85E, A95 = 6.1 (Hodych & Hayatsu 1988) (b) 63N, 85E, A95 = 7.0 (This study). RPD—relative palaeolatitude displacement, positive is northward, negative is southward. RR—relative rotations about vertical axes, positive is anticlockwise, negative is clockwise. *—RPD was calculated using average inclinations and RR is reported for each of three study areas.

5.3 Tectonic analysis

The Early Jurassic reference pole of Hodych & Hayatsu (1988) and a combined Colorado–Newark reference pole calculated in this study (Table 4) have been used in the following tectonic analysis. Using the tectonic analysis outlined by Yole & Irving (1980) and incorporating the error analysis of Demarest (1983), the Hazelton Group remanences are latitudinally concordant within statistical error with either the combined Colorado–Newark reference pole (this study) or the Newark reference pole (Hodych & Hayatsu 1988) (Table 5). This result is consistent with our re-evaluation of Triassic and Early Jurassic palaeomagnetic results that previously supported the ~1300 km northward displacement hypothesis (Table 5). This result is also consistent with data from the Permian Asitka volcanic rocks from Stikinia that Irving & Monger (1987) have interpreted as latitudinally concordant. The average magnitude of northward displacement for the seven studies in Table 5 is 1° ± 9° for both the accepted reference poles. Such precision is beyond that of any single palaeomagnetic study and is for all intents and purposes equal to zero. This analysis indicates that Terrane I, from the Permian to Early Jurassic, was in its present latitudinal position relative to the craton, although it has undergone large and variable rotations especially near the margins of the Stikine Terrane. Similar latitudinal concordance of southern Wrangellia (Vancouver Island) with Terrane I and North America is suggested by the single Bonanza Group result from Vancouver Island (Table 5). The data from the Karmutsen Formation (Symons 1971; Schwartz, Muller & Clark 1980; Yole & Irving 1980), also on Vancouver Island, have not been included in this analysis as they appear to have a very complex magnetization history and the results are somewhat ambiguous.

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5.4 Geotectonics

Terrane I, an allochthonous microplate, is thought to have been formed by the assembly of four subterranes in the Middle Triassic (Monger *et al.* 1982). The concordant palaeolatitudes obtained from the Stikine and Quesnel subterranes in Terrane I is consistent with this hypothesis, although the large and variable internal rotations suggest that the assembly of Terrane I was not complete by the Early Jurassic. It is suggested that southern Wrangellia (Vancouver Island), part of the outboard Insular belt, also characterized by large rotations and latitudinal concordance from the Bonanza Group data (Table 5), was also close to its present position relative to Terrane I and North America. Recent palaeomagnetic studies on Middle Eocene volcanics from Stikinia and Quesnellia, (Symons & Wellings 1989; Bardoux & Irving 1989; Vandall & Palmer 1990) show that Terrane I and much of the Canadian Cordillera had assembled and docked with North America by at least Middle Eocene. Thus Terrane I must have been stabilized by at least the Middle Eocene because the main body of the Canadian Cordillera palaeopoles from younger rocks in the Cordillera are concordant.

Cordilleran Cretaceous palaeomagnetic data remain largely discordant (e.g. Beck *et al.* 1981; Irving *et al.* 1985; Rees, Irving & Brown 1985; Butler, Harms & Gabrielse 1988; Marquis & Globerman 1988), and are interpreted to indicate that much of the western Canadian Cordillera was situated as far as 2000 km south of its present position at that time. Geologic evidence suggests that the various terranes of the western Canadian Cordillera were amalgamated by the mid-Cretaceous (Monger *et al.* 1982). This is supported by the Cretaceous palaeomagnetic data. Although discordant, Irving *et al.* (1985) suggest their vertical axis rotations are systematically ~60° clockwise which is in contrast to the Permian to Early Jurassic results, indicating that the various terranes of the western Canadian Cordillera had assembled as a unit by this time. The latitudinally concordant data both before and after the Cretaceous constrains the timing of large-scale displacements within this period. Displacement models based on the discordant Cretaceous data must accommodate large-scale southward displacement of western parts of the Canadian Cordillera after the Early Jurassic, followed by northward displacement with 60° clockwise rotation, and then stabilization of the Canadian Cordillera by Middle Eocene. Umhoefer (1987) developed a displacement model, based for the most part on palaeomagnetic evidence, which is interpreted to be consistent with the known geology of the reconstructed margin of western North America. It can accommodate up to ~2000 km of northward displacement of 'Baja British Columbia' (Insular and Intermontane composite terranes which are roughly comparable to Terranes I and II) as part of the Kula plate along a transform margin with North America between 85 and 66 Ma. However, clockwise rotation is limited to 20° (Umhoefer 1987) which conflicts with the inferred ~60° clockwise rotation (Irving *et al.* 1985) observed in mid-Cretaceous palaeomagnetic results. Also the model does not consider how 'Baja British Columbia' was first displaced to the south from its latitudinally concordant Permian to Early Jurassic palaeolatitudes. As pointed out by Butler *et al.* (1989) the magnitude and timing of this displacement is inconsistent with the currently perceived pre-mid-Cretaceous links between the North American miogeocline and parts of western Cordillera (Price & Carmichael 1986). Symons (1977), Beck *et al.* (1981) and Irving *et al.* (1985) indicated that mid-Cretaceous palaeomagnetic discordancy could also be explained by regional tilting of 30° to the SW. Recently Butler *et al.* (1989) have advanced this hypothesis with stratigraphic, petrologic and isotopic data that indicate that parts of the Coast Plutonic Complex and the North Cascade Range have been tilted by ~30° to the SW. Certainly the regional tilt hypothesis appears more reconcilable in view of the latitudinal concordancy of the Permian to Early Jurassic and Middle Eocene to Recent data. However, this hypothesis cannot account for discordant Cretaceous data from bedded volcanic rocks in Terrane I (e.g. Marquis & Globerman 1988). It is therefore probable that Cretaceous Cordilleran tectonics were characterized by both regional tilting in outboard terranes and moderate latitudinal displacement in the inboard terranes. Certainly many further palaeomagnetic studies of Cretaceous bedded rocks from the Canadian Cordillera are needed to resolve the tilt versus displacement controversy.

6 CONCLUSIONS

The Early Jurassic Telkwa Formation of the Hazelton Group is characterized by a stable, two polarity, pre-folding, primary remanence that was acquired upon cooling and deposition of volcanic lavas and tuffs. These palaeomagnetic results confirm the earlier but less precise results of Monger & Irving (1980). Palaeomagnetic results from the Permian to Early Jurassic are consistently characterized by large and variable clockwise and anticlockwise rotations about vertical axes, indicating that amalgamation of the large composite Terranes I and II was not yet complete. Because of the downward revision of the geologic time-scale and changes to the Early Jurassic reference pole for North America, Terrane I is no longer required to have been moved ~1300 km from the south since the Early Jurassic and was, in fact, close to its present latitudinal position with respect to North America. The interpretation of results from studies which were consistent with Monger & Irving's (1980) conclusion are shown to be latitudinally concordant with the expected Early Jurassic reference pole for North America. Given that palaeomagnetic data from the Permian to Early Jurassic and Middle Eocene to Recent are latitudinally concordant, displacement models based on the discordant Cretaceous data must accommodate large-scale southward displacement of western parts of the Canadian Cordillera after the Early Jurassic. This must be followed by northward displacement with 60° of clockwise rotation, and finally stabilization of the Canadian Cordillera by the Middle Eocene.

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